

Efficient Audiovisual Speech Processing via MUTUD: Multimodal Training and Unimodal Deployment

Hypothesis: A model can be trained using multiple modalities (e.g., audio and video) but deployed using only a single modality by using a novel Temporally Aligned Modality feature Estimation (TAME) module to estimate the features of the missing modality during inference. This approach is expected to yield a unimodal model that performs better than a standard model trained on single-modality data, while being more computationally efficient at inference time than a full multimodal model.

Related Work: Multimodal speech processing systems, particularly audiovisual (AV) ones, often outperform unimodal systems, especially in noisy conditions. However, they suffer from practical constraints like high computational cost, complex sensor requirements, and the need for synchronized multimodal data during inference, which limits their real-world deployment. While methods for resource-constrained learning and handling missing modalities exist, they are often not designed for the specific scenario of unimodal deployment with efficiency as a primary goal. For instance, self-supervised frameworks like AV-HuBERT are large and not easily adapted for tasks like speech enhancement. This paper addresses the gap by proposing a method to explicitly leverage multimodal data during training to create a highly efficient and performant model for unimodal deployment.

Abstract: [Content omitted for brevity]

Experiments:

Description: To evaluate the MUTUD framework on the task of audiovisual speech enhancement (AVSE) under various noise conditions and compare its performance and efficiency against unimodal and multimodal baselines.

Method: The experiment will use the LRS3-TED dataset for speech and the DNS Challenge dataset for noise. The task is audiovisual speech enhancement. The proposed MUTUD model, based on a GCRN U-Net architecture with a ResNet18 video encoder, will be compared against three baselines: a standard audio-only model, an audio-only model with a matched number of parameters to MUTUD, and a full audiovisual model. Two training strategies for MUTUD will be explored: training from scratch and pre-training the TAME module.

Metrics: Short Time Objective Intelligibility (STOI), Scale-Invariant Signal-to-Distortion Ratio (SISDR), Perceptual Evaluation of Speech Quality (PESQ)

Description: To demonstrate the effectiveness of the MUTUD framework on the task of audiovisual speech recognition (AVSR), particularly in low-SNR scenarios;

Method: The experiment will use the LRS3-TED dataset. The task is audiovisual speech recognition. The MUTUD framework will be integrated into the V-CAFE baseline architecture. The performance of the resulting model will be compared against an audio-only baseline and the full audiovisual V-CAFE model.

Metrics: Word Error Rate (WER)

Description: To showcase the versatility of the MUTUD framework by applying it to audiovisual active speaker detection (AV-ASD) in a scenario where the audio modality is absent at inference time (i.e., video-only deployment).

Method: The experiment will use the challenging, ego-centric EasyCom dataset. The task is active speaker detection. A network with 3D convolutional layers and an LSTM will be designed. The TAME module will be used to estimate audio representations from video inputs during inference. The proposed MUTUD model will be compared against a video-only baseline and a full audiovisual baseline.

Metrics: mean Average Precision (mAP)

Risk Factor:

While the MUTUD introduces a novel approach to multimodal training with the proposed TAME module and applies it to audiovisual speech tasks, its behaviour on other tasks and models in multimodal domain remains to be seen.

So further exploration through even more complex tasks and models may shed more light on MUTUD.

Meta ICLR criticism: The reviewers have **primarily expressed concerns regarding the experiments** presented in the paper. While the paper includes a series of ablation studies, the main issues highlighted involve **unfair comparisons** and the **absence of certain ablations and analyses**. Additionally, some reviewers have noted **a lack of novelty**, pointing out that **similar methods have already been introduced**. Although the authors have submitted additional results and explanations to bolster their case, **key issues, such as the unfair comparisons and the novelty of the method, remain unresolved**

Mean ICLR
Soundness Score
1.75/4.0

